



Masters in Business Analytics
MSBA 502-01: Analytics Programming I
Syllabus, Fall 2026

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Course Logistics

Course Meeting Room: KCBE 328

Course Dates: Mondays and Wednesdays

Course Time: 2:30 – 5:20 pm

Instructor Office Hours:

- In-person: Tuesdays: 2:30 – 4:30 pm
- Zoom: By appointment. Notify me early if you want to use Zoom for office hours

Office Hours Zoom Link: <https://san Diego.zoom.us/j/98756496792>

Course Description

While business analytics is often perceived as the process of generating insights from data through various analytical methods, programming plays a pivotal role in this field, underpinning the entire analytics landscape. It empowers analysts to collect, store, clean, and transform data, conduct in-depth analyses, automate repetitive tasks, and visualize both data and results. The versatility of programming languages makes them indispensable for a wide range of analytics projects. Indeed, most analytics roles—and many non-analytics roles—demand proficiency in at least one programming language, even in the age of artificial intelligence (AI).¹ Industry professionals frequently utilize languages like Python, R, and SQL, among others, depending on the specific objectives of their projects.

This course will focus on leveraging Python for analytics. Python is a powerful, open-source programming language known for fostering collaboration and innovation. Its open nature has led to the development of numerous libraries available online that support data analysis, which you can adapt and incorporate into your own code. Python's popularity is also attributed to its simple syntax, which makes it easy to collect, filter, and analyze datasets of various sizes, and to build and test models ranging from simple linear regressions to advanced neural networks.

We will use Jupyter Notebook for Python programming. You can download and install Anaconda Navigator (free to download and use) and use Jupyter Notebook from it (the simplest way). I have uploaded a video and a separate file with download instructions

¹ See <https://www.forbes.com/sites/amirhusain/2025/10/21/programming-in-the-age-of-ai-why-code-still-matters/>

(for Windows; Mac will not be very different). They will also contain information about using virtual environments for running your Python code. **Please ensure you arrive at class with a fully configured and running Python Jupyter Notebook environment.**

The course is created more from a data analysis perspective than a software development one. In that regard, emphasis is placed on different aspects of data analysis in practice. We are often interested in solving a business problem by using data, instead of just writing/creating blocks of code. It is expected that the course will help students expand their analytics competencies in areas such as understanding business problems, converting these problems into technical structures, and solving them using data and Python. Analytics systems (and codes) developed by humans can be subject to bias and judgment. Analytics professionals should learn to evaluate the ethical and social implications of these biases and ensure that the technology is used in a productive and positive manner.

Course Learning Outcomes

- Develop expertise in Python to analyze data, interpret results, and visualize insights
- Apply appropriate AI/ML techniques to important business problems
- Generate insights by using regression, classification, and clustering methods
- Develop skills in several Python libraries, such as Pandas, Matplotlib, NumPy, Statsmodels, Scikit-learn, and Seaborn

Course Materials (Required)

- There is no required textbook for this course. There are several thousand resources available online (free). Feel free to refer to the supplemental resources.

Supplementary Resources:

- An Introduction to Statistical Learning with Applications in Python by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor (Free version available online)
- Deep Learning by Ian Goodfellow and Yoshua Bengio and Aaron Courville (Free version available online)
- [Python for Everybody](#)
- [Python Data Science Handbook](#)

Practice:

Coding practice is often considered the most crucial element in gaining mastery in Python (or any programming language). The hands-on experience gained through practice is irreplaceable and is what truly differentiates a novice from an expert. Thus, my suggestion is: PRACTICE....PRACTICE....PRACTICE....

- Excellent for beginners: <https://www.w3schools.com/python/>
- Great for preparation for big tech companies: <https://leetcode.com/>
- Coding exercises: <https://www.dataquest.io/blog/python-practice/>
- And many more on YouTube. Do not just watch anyone code; do it.

Datasets:

Finding datasets is a crucial task (especially for your project). You can explore these following repositories. There are more resources, obviously.

- <https://www.kaggle.com/>
- <https://www.tableau.com/learn/articles/free-public-data-sets>
- <https://data.gov/>
- <https://libraryguides.missouri.edu/datasets/public-use>
- <https://archive.ics.uci.edu/datasets>
- <https://data.worldbank.org/>

Classroom Behavior

- MARKET YOURSELF to your classmates and professor. Hone your professional conduct skills now to prepare for a successful future. This includes but is not limited to:
 - *putting away your computer when asked*
 - *turning off/muting cell phones before entering the classroom*
 - *arriving on time and staying for the entire duration*
 - *being prepared for class and not engaging in conversations during class time, which can disrupt the class discussions*
 - *paying attention during class*
 - *do not engage in any academic dishonesty, which may include the misuse of AI tools*
- ACTIVELY PARTICIPATE during each session. Feel free to ask questions during class. Please do not feel shy or embarrassed; if you are unclear on an idea, the chances are high that many of your classmates are too. However, only one person in the class should speak at one time. Please be respectful to others and raise your hand if you have questions or comments during the lecture.
- USE YOUR COMPUTER/LAPTOP EFFECTIVELY for classroom purposes. Avoid using computers for non-academic purposes such as shopping or browsing unrelated websites. Mute notifications and keep the sound off to avoid disruptions. Follow all institutional policies regarding technology use. Be mindful of intellectual property and copyright laws. The default policy is not to use AI on any assignments unless mentioned otherwise. **While AI can be a powerful tool to support your learning, it should complement—not replace—your own effort and understanding of the material.**
- READ THE ASSIGNED ARTICLES AND LECTURE MATERIALS before coming to the class. In this classroom, we will engage in discussions to explore various ideas, offer constructive critiques, and collaboratively identify best practices.
- TREAT EACH CLASS AS A SKILL DEVELOPMENT AND PRACTICE SESSION. Work on your code, clear any doubts, and explore ways to collect, clean, and analyze data.

Course (Instructor) Evaluations

Near the end of this course, you will have access to an online evaluation. Your timely and considerate feedback is valuable to us and an important element of your learning experience.

Academic Honesty Statement

Acts of academic dishonesty are taken seriously and dealt with harshly. In preparing for class or assignments, **you must not use notes** from another section of this course, any previous offering of this course, or a similar course, or discuss the material with students who have already taken the course or your current classmates. This restriction extends to simulations and case-related information obtained from other sources. *Use of ChatGPT, Gemini, or any other AI program is not recommended unless explicitly allowed for this course, since this is a foundational course in learning Python for data analysis.* You can use the websites suggested by Google Search to develop your code or seek any solutions to resolve coding issues; you are not recommended to let AI write and/or execute your code.

In addition to the points above, note that the University of San Diego policy on the Integrity of Scholarship is an integral part of this course. Violations of academic integrity include the following: (1) unauthorized assistance on assignments or exams; (2) falsification or invention of data; (3) unauthorized collaboration on any assignment or exam; (4) plagiarism (you must cite the source of any material/sentence/paragraph that does not originate from you); (5) misappropriation of research materials; and (6) any unauthorized access of an instructor's files or computer account. Further detail is provided via this link to the USD policies: <http://catalogs.sandiego.edu/graduate/academic-regs/>.

AI Use Policy

The Knauss School of Business at the University of San Diego promotes responsible classroom use of AI through approved tools such as Google Gemini, available to students, faculty, and staff for brainstorming, coding, and research. USD emphasizes ethical AI integration, prohibits unauthorized generative AI use in coursework unless permitted by instructors, and provides AI training through professional education programs. I will be very clear about when and how AI will be used in the classroom.

Now, more than ever, there is easy access to assignment assistance. It can be tempting to use these resources. This is not the time to use them. Imagine being back in 3rd and 4th grade, working on division, multiplication, and fractions, and some of your classmates get to use a calculator when everyone else must work by hand. This doesn't make sense at that moment because of an imbalance among students. It doesn't make sense in the long run because knowing how to complete that math by hand helps with higher-level math. My job is to work hard to ensure everyone in our class has the resources needed to succeed in this course. It is also my job to be diligent about fairness.

At the same time, this course will develop ethical and effective analytics skills. The goal is not to substitute your thinking, creativity, or effort with AI, but to learn how to amplify your capabilities through responsible use of these tools. Strong professionals will not simply rely on AI to produce answers; they will know how to guide, evaluate, refine, and improve AI-generated outputs using their own expertise and judgment. When used effectively, AI can increase efficiency, deepen analysis, improve communication, and enhance creativity. Your ideas, critical thinking, and domain knowledge combined with AI's computational power can produce work that is stronger than either could achieve alone. The emphasis in this course will therefore be on learning how to work in sync with AI tools while continuing to develop the foundational skills and independent thinking that remain essential for long-term success. *We will extensively use AI in MSBA 503.*

If you get stuck at any point this semester, let me know...

Disability Statement

The University of San Diego complies with the Americans with Disabilities Act, and Section 504 of the Rehabilitation Act. Any student who may require an accommodation under such provisions needs to formally register and provide documentation of his/her disability to Disability and Learning Difference Resource Center, Serra Hall Room 300, phone: [\(619\) 260-4655](tel:619-260-4655), email: disabilityservices@san Diego.edu. Once registered, the student will need to obtain a disability verification memo from the Disability and Learning Difference Resource Center office, and provide a copy of this memo to the course instructor (**as soon as possible, ideally at the beginning of the term**). Please note that your instructor is unable to provide any disability-related accommodations unless he receives this disability verification memo, and no retroactive accommodations can be made.

Late Policy

All assignments and problem sets should be uploaded to Canvas. You do not need to submit the project presentation to Canvas. Late submissions are not accepted except for excruciating circumstances (why delay work when you can submit it early and relax).

Communications

You are highly encouraged to communicate constantly and openly with me about any problems you may have or feedback for the course. The preferred channel is through email. Please check your email and Canvas regularly. We will use these ways to communicate important course news. **Include MSBA:502 in the subject line of your email.**

Course Evaluation

Everyone is expected to attend and to prepare fully for each day. Class participation involves (a) attendance, (b) classroom citizenship, and (c) **your engagement with the class materials, including coding practice**. Do not be afraid to make points that you may regard as minor, ask clarifying questions, or otherwise contribute in small ways. Essential preparation includes reading the assigned readings and cases, doing the assignments, and bringing these resources and materials to each class.

Please be on time for each class. Failure to attend class can adversely affect your grade. If you miss a class for any university excused absences, let me know in advance in writing. As a rough guideline, the following describes examples of behaviors that are associated with different class participation grades:

A: Exceptional – Always in class; works on the materials actively and consistently in every session and helps the class to be successful.

B: Good – Almost always in class (no more than 1 absence) and working on the exercises; occasionally (i.e., not consistently) contributes to class discussions.

C: Average – Almost always in class (no more than 1 absence); primarily an observer in class and contributes little to the learning environment of the class.

D: Weak – Frequent (2 to 3) absences; makes little to no contribution when in class and does not work on the materials.

F: Poor – Persistent absence (exceeding 3 absences over the semester), and makes little to no contribution when in class. Use of a laptop, tablet, or cell phone in class results in an F grade for that day. Absences of 25% or more (7 or more) of the classes will result in a grade of “F” for the course.

Student Assessments

Each student’s final numerical score for the course is based on the following items and weights:

Participation	10%
Problem Set	10%
Assignments (2 total, 15% each)	30%
Case Study	10%
Midterm	20%
Final Project	20%
Total	100%

Grading Inquiries:

All grading inquiries must be type-written and made within 2 days of the date the graded exam or assignment is returned to class (regardless of your attendance). It is your responsibility to make sure that all grades have been accurately recorded on Canvas. If a grade has not been recorded correctly or is missing from Canvas, you must notify the professor BEFORE the last day of class, or your final grade will be calculated with the incorrect or missing score. Inquiries about your final grade must be made prior to August 30. Inquiries brought after the deadline will not be considered. Out of fairness to your classmates, there will be no regrades of any assignments. If you are concerned about your grade, please come speak to me as soon as possible in the semester. Please note that under the University of San Diego policy, your grade for the course may only be changed if a clerical error has been made.

Grading Schema

Total Grade	Letter Grade
93% and above	A
90% and below 93%	A-
87% and below 90%	B+
83% and below 87%	B
80% and below 83%	B-
77% and below 80%	C+
73% and below 77%	C
70% and below 73%	C-
67% and below 70%	D+
63% and below 67%	D
60% and below 63%	D-
Below 60%	F

Course Outline: Here is an outline of the topics for this course. Please note that it can change. However, given that this is a fast-paced course, I will try my best to keep it as it is. All assignments are due by the end of the day (11:59 pm Pacific time). Deadlines are always 11:59 pm Pacific time. Problem Set 1 is not due on a regular class day; a little bit of extra time is provided since it is the first assignment of the course, and a little more time is never bad for that.

Day	Lecture Module	Topic(s)	Other Readings/Practice	Assignment Due
Sep 1 (Special case) Tuesday (Aug 31's class)	1	Course introduction, syllabus, brief introduction to AI, ML, and DL, introduction to Python	Python for Everybody	
Sep 2 Wednesday	1	Data types, errors, modular programming		
Sep 7 Monday	No class – Labor Day Holiday			
Sep 9 Wednesday	1	Data types, errors, modular programming contd.	W3schools – complete the Python Home section (at least)	Problem Set submission (Take Home) Friday Sep 11 EOD
Sep 14 Monday	2	Language of data (Review of statistical concepts, probability, hypothesis testing), and Pandas		GitHub Assignment (before class)

Day	Lecture Module	Topic(s)	Other Readings/Practice	Assignment Due
Sep 16 Wednesday	2	Pandas continued, Matplotlib	Related W3schools sections	
Sep 21 Monday		In-class assignment		
Sep 23 Wednesday	2	NumPy, Categorical data, Seaborn	Related W3schools sections	Take Home Assignment Friday Sep 25 EOD
Sep 28 Monday	3	Linear Regression and Statsmodels library	ISLP Chapter 3	
Sep 30 Wednesday		Mid-term (In-class)		
Oct 5 Monday	4	Machine Learning Concepts, Logistic Regression, and Scikit-Learn library		
Oct 7 Wednesday	4	Case study discussion Clustering		Case Study Submission Before Class
Oct 12 Monday		Project Work (No regular class)		Project Team and One-Page Draft Submission Before Class
Oct 14 Wednesday		Presentations		
Oct 19 Monday		Presentations		Submit your Project Report
Oct 21 Wednesday		Project and course end discussions and brief conversations about other learning approaches		

Deliverables

- Problem Set:** The problem set is created to test your basic programming skills in Python. The problem set will present a scenario where you need to understand the context, make realistic assumptions, generate appropriate data, and follow logic. Do not use any AI tool to generate your code; however, you can use it for any contextual information. You need to cite them in that case, or any articles that you

use from online sources. **This is an individual assignment. However, you may seek help from your peers. If you do that, mention it in the acknowledgment section. It will not impact your grade. However, you cannot submit others' work or work jointly with someone.** Submit a Jupyter Notebook file with proper comments. The code and corresponding outputs should be present when I open your Jupyter file. Your code should run fully without any issues and produce the outputs for full points.

- **Assignments:** The assignments require more effort than the problem sets. However, as you practice more, you are expected to become more comfortable with coding. Also, you will be able to use Python's libraries, which will substantially reduce your coding efforts. **Do not use any AI tool to generate your code unless explicitly mentioned.** Unlike the problem set, assignments are designed to be more straightforward; you do not need to make any assumptions or need any contextual information. You need to cite them in that case or any articles that you use from online sources. **These are individual assignments. However, you may seek help from your peers for the take-home assignment. If you do, mention it in the acknowledgment section. It will not impact your grade. However, you cannot submit others' work or jointly with someone. One assignment needs to be completed in class without any help,** and one will be take-home. Submit the Jupyter notebook file, as you did for the problem set.
- **Mid-term:** The mid-term will be a dataset-based assignment. You may use any AI tool; however, you cannot simply feed the dataset into an AI tool and generate code and results. You can use an AI tool as a guide. However, you are responsible for the accuracy of any work you submit, including work completed with the assistance of AI tools. AI-generated code, analyses, results, and explanations may contain errors. Therefore, I strongly recommend verifying your work using multiple sources, such as course materials, documentation, textbooks, and independent calculations or tests. Any errors that remain in the submitted work are ultimately your responsibility. You can not seek help from anyone else.
- **Case Study: Data Science at the Warriors:** Prepare the Data Science at the Warriors case from the HBR coursepack. Read this case carefully for classroom discussion and answer the following questions. Upload them (Canvas) before class on October 7. Each question carries 4 points.
 1. Why did the Warriors invest in data science? Please explain all the factors briefly.
 2. Comment on the decision to have the data science team under marketing.
 3. Should they delay by another year for testing or deploy in the current season? Comment.
 4. Explain the outcome variables highlighted in the case and their importance.
 5. Conduct the analysis for the days before game, as described in the case based on the dataset provided.
- **Project:** You can work on the final project **by working with one (recommended) or, at most, two more peers.** The idea is to develop a project that will have a problem statement and will focus on data collection, cleaning and transformation, analysis, visualization, and recommendations. While I intend to keep the flow of

the presentation going, I may ask you to explain a part of your code during the presentation. Note that you can be more strategic in choosing your project. In the Advanced Analytics for AI class, you will explore more advanced topics, including web scraping, text and image analysis, among others. Thus, you can choose a topic for the project for this class that you can also extend with more data and/or analysis in that class. Otherwise, a completely new project is also acceptable. I have included a file on project expectations.